	MIDTERM EXAMINATION SEMESTER FALL 2003 CS301-DATA STRUCTURE	Total Marks:86 Duration: 60min

Instructions

Please read the following instructions carefully before attempting any question:

1. **The duration of this examination is 60 Mins.**
2. **This examination is closed book, closed notes, closed neighbors; any one found cheating will get no grade.**
3. **Unless stated otherwise, all questions carry a single mark.**
4. **Do not ask any questions about the contents of this examination from anyone.**
    a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
    b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
5. **Most, but not all, of the examination consists of multiple-choice questions. Choose only one choice as your answer.**
    a. **If you believe that two (or more) of the choices are the correct ones for a particular question, choose the best one.**
    b. **On the other hand, if you believe that all of the choices provided for a particular question are the wrong ones, select the one that appears to you as being the least wrong.**
7. **You are allowed to use any development environment like Dev C++ etc.**

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Question No: 1 **Marks: 2**

Given the class declaration

```
class MyClass
{
      public:
      ...
      void Func();
      private:
      int n;
};
```

what notation does the body of `Func` use to assign the value `3` to `n`?

(a) n = 3;
(b) MyClass.n = 3;
(c) MyClass::n = 3;
(d) someObject.n = 3;
(d) It can't be done–n is private.

Question No: 2 **Marks: 2**

Suppose that the class declaration of `SomeClass` includes the following function prototype.

```
bool LessThan( SomeClass anotherObject );
```

Which of the following tests in the client code correctly compares two class objects `alpha` and `beta`?

(a) if (alpha < beta)
(b) if (alpha.LessThan(beta))
(c) if (LessThan(alpha, beta))
(d) if (alpha.LessThan.beta)
(e) if (LessThan(alpha).beta)

Question No: 3 **Marks: 2**

When should you use a const reference parameter?

(a) Whenever the data type might be many bytes.
(b) Whenever the data type might be many bytes, the function changes the parameter within its body, and you do NOT want these changes to alter the actual argument.
(c) Whenever the data type might be many bytes, the function changes the parameter within its body, and you DO want these changes to alter the actual argument.
(d) Whenever the data type might be many bytes, and the function does not change the parameter within its body.

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Question No: 4	Marks: 2

The Bag ADT is like the List ADT. The Bag ADT does not store items in any particular order and it allows duplicates. Suppose that the `Bag` class is efficiently implemented with a fixed array with a capacity of 4000. Insert appends the new item at the end of the array. Choose the best description of `b`'s member variables `size` (count of items in the bag) and `data` (the array that holds the actual items) after we execute these statements:

```
Bag b;
b.insert(5);
b.insert(4);
b.insert(6);
```

What will be the values of b.size and b.data after the statements?

(a) b.size is 3, b.data[0] is 4, b.data[1] is 5, b.data[2] is 6
(b) b.size is 3, b.data[0] is 5, b.data[1] is 4, b.data[2] is 6
(c) b.size is 3, b.data[0] is 6, b.data[1] is 4, b.data[2] is 5
(d) b.size is 3, b.data[0] is 6, b.data[1] is 5, b.data[2] is 4

Question No: 5 **Marks: 2**

The operation for removing an entry from a stack is traditionally called:

(a) delete
(b) peek
(c) pop
(d) remove

Question No: 6 **Marks: 2**

Which of the following stack operations could result in stack underflow?

(a) is empty
(b) pop
(c) push
(d) Two or more of the above answers

Question No: 7 **Marks: 5**

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Consider the following pseudo code:

```
declare a stack of characters
while ( there are more characters in the word to read )
{
     read a character
     push the character on the stack
}

while ( the stack is not empty )
{
     pop a character off the stack
     write the character to the screen
}
```

What is written to the screen for the input "carpets"?

(a) serc
(b) carpets
(c) steprac
(d) ccaarrppeettss

Question No: 8 **Marks: 2**

In the linked list implementation of the stack class, where does the push member function place the new entry on the linked list?

(a) At the head
(b) At the tail
(c) After all other entries that are greater than the new entry.
(d) After all other entries that are smaller than the new entry.

Question No: 9 **Marks: 2**

Given a stack of n items, how many POP and PUSH operations need to be performed to remove the item at its bottom?

(a) 0 POP operation and 0 PUSH operation
(b) 1 POP operation and 1 PUSH operation
(c) n POP operations and n PUSH operations
(d) n POP operations and n-1 PUSH operations
(e) Unknown

Question No: 10 **Marks: 2**

In the linked list implementation of the queue class, where does the insert member function place the new entry on the linked list?

(a) At the head
(b) At the tail
(c) After all other entries that are greater than the new entry.
(d) After all other entries that are smaller than the new entry.

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Question No: 11	Marks: 2

I have implemented the queue with a linked list, keeping track of a front pointer and a rear pointer. Which of these pointers will change during an insertion into a *NONEMPTY* queue?

(a) Neither changes
(b) Only front pointer changes.
(c) Only rear pointer changes.
(d) Both change.

Question No: 12 **Marks: 2**

I have implemented the queue with a linked list, keeping track of a front pointer and a rear pointer. Which of these pointers will change during an insertion into an *EMPTY* queue?

(a) Neither changes
(b) Only front pointer changes.
(c) Only rear pointer changes.
(d) Both change.

Question No: 13 **Marks: 6**

For public part of the Throttle declaration below, mark each function member header as follows:

- **Mark C for any constructor;**
- **mark X for any function that is forbidden from changing the throttles data fields.**

```
class Throttle
{
     public:
     Throttle( );
     Throttle(int size);
     void shut_off( );
     void shift(int amount);
     double flow( ) const;
     bool is_on( ) const;
     ...
```

Answer/Solution

```
class Throttle
{
     public:
     Throttle(     );     C
     Throttle(int size); C
     void shut_off( );
     void shift(int amount);
     double flow( ) const;     X
     bool is_on( ) const;      X
     ...
```

Question No: 14 **Marks: 5**

I am going to execute this code with THREE pushes and ONE pop:

```
Stack s;
s.push(1);
s.push(2);
s.push(3);
cout << s.pop( );
```

Suppose that the stack `s` is represented by a *singly linked list*. Draw the linked list after the above operations and show where the top element is in the list.

```
head -->
```

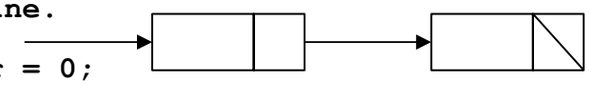
Answer/Solution

head 2 1

Question No: 15 **Marks: 10**

Complete the body of this function. Use a Queue of characters to store the input line as it is being read.

```
int counter( )
// Precondition:
// There is a line of input waiting to be read from cin.
// Postcondition:
// A line of input has been read from cin, up to but not
// including the newline character. The return value of
// the function is the number of times that the LAST
// character of the line appeared somewhere in this line.
// EXAMPLE
// Input: PQQYDYYTY
// The value returned by the function counter would
// be 4 for this input since there are 4 Y's in
// the input line.
{
     int answer = 0;
     Queue q;
```



Answer/Solution

```
int counter()
{
   char a[100];
   int i=0;
   int answer=0;
   Queue q;

   cin.getline(a,98,'\n');
   for(i=0;i<strlen(a);i++)
   {
      q.enqueue(a[i]);
   }

   i--;
   while(!q.isEmpty())
```

```
    {
      if(a[i]==q.dequeue())
      {
        answer++;
      }
    }
    return answer;

}
```

Question No: 16 **Marks: 5**

. I am going to execute this code with THREE inserts (enqueue) and ONE remove (dequeue):

```
Queue s;
s.insert(1);
s.insert(2);
s.insert(3);
cout << s.remove( );
```

Suppose that queue `s` is represented by a circular array. Draw the state of the private member variables "`data`" and "`front`" of `s` after the above code:

```
          0   1   2   3   4   5   6   7   8   9
        +---+---+---+---+---+---+---+---+---+---+          +---+
data:   |   |   |   |   |   |   |   |   |   |   |   front: |   |
        +---+---+---+---+---+---+---+---+---+---+          +---+
```

Answer/Solution

0	1	2	3	4	5	6	7	8	9
	2	3							

Front 1

Question No: 17 **Marks: 10**

Consider `CList` and `Node` classes defined as follows:

```
class Node

{
      public:
      Node *next;
      Node *prev;
      int data;
};
class CList
{
      public:
      void insertHead(int);
      void insertTail(int);
      void removeHead();
      void removeTail();
      bool isEmpty();
      bool find(int);
      private:
      Node *head;
      Node *tail;
};
```

A. write the body of the member function `insertHead` which inserts a new element at the head of the list.

```
void Clist::insertHead( int x )
{
```

B. write the body of the member function `removeTail` which removes the element at the tail of the list.

```
void Clist::removeTail( int x )
{
```

Answer/Solution

(a) Solution for Question 17 option (a)

```
void CList::insertHead(int x)
{
    Node *newNode=new Node();
    newNode->data=x;
    newNode->next=NULL;
    newNode->prev=NULL;
    if(isEmpty())
        head=tail=newNode;
    else
    {   newNode->next=head;
        newNode->prev=NULL;
        head->prev=newNode;
        head=newNode;
    }
}
```

(b) Solution for Question 17 option (b)

```
void CList::removeTail(int &x)
{
    if(isEmpty())
```

```
        return;
    else
    {
        Node *p=tail;
        if(head==tail)
                head=tail=NULL;
        else
        {
                tail=tail->prev;
                tail->next=NULL;
        }
        x=p->data;
        delete p;
        return;
    }
}
```

Question No: 18 **Marks: 10**

Trace the running of the infix to postfix conversion algorithm on the infix expression
A * (B − C)/D

symbol	postfix string	operator stack

A * (B – C) / D

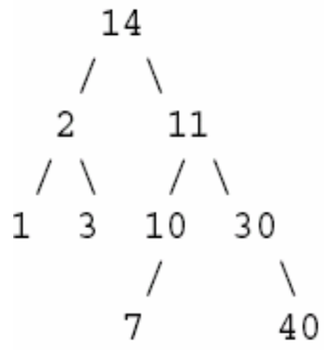
Symbol	Postfix	Stack
A	A	
*	A	*
(	A	*(
B	AB	*(
-	AB	*(-
C	ABC	*(-

	)	ABC- ABC-	*(*

/	ABC-*	/
D	ABC-*D	
	ABC-*D/	

Question No: 19 **Marks: 13**

Here is a small binary tree:



A. What are all the leaves? (2pts)
C. What are the ancestors of the node 30?(2pts)
D. What are the descendants of the node 11? (2pts)
E. Is the tree a binary search tree (BST) (true/false)? (2pts)
F. Print the tree when visited in in-order manner? (5pts)

Answer/Solution

A) Leaves of the Tree = 1,3,7,40
B) Ancestors of the node 30 = 11,14
C) Descendants of the node 11= 10,30,7,40
D) Is the three a binary search tree (BST) (True/False) False
E) In-order Traversal = 1,2,3,14,7,10,11,30,40